Reliable Team Assistant: Phase 1

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Question and implemented scope

Can dependency-aware recovery preserve evidence-backed diagnosis quality with less recomputation after a trusted invalidation event? Configuration and telemetry roles collect read-only snapshots; a coordinator diagnoses. The runtime records consumed versions, invalidates dependent work, retains independent artifacts, rejects obsolete publications and links corrections. Missing recorded provenance triggers restart in development controls; this fallback was not exercised in the formal batch. Arbitrary error discovery, operational writes and invisible dependencies are outside the implemented guarantee.

Workload and frozen protocol

Cloud-OpsBench is pinned to ea05dafbc87b5985df8721622d63d4229c2b4224. Formal cases boutique/runtime/32, service/12 and performance/5 each run twice, reversing restart/selective order on repeat 2. These are three independent cases, not six. TritonAI api-glm-5.3 uses low reasoning, temperature 0 and 8192 output tokens. Course DataHub CPU 2/8 GB runs isolated Python 3.11.9 with exact dependency pins. Source 1380c6b identifies execution; the complete freeze is archived.

The user advanced launch to October 24 before formal observations were inspected. Cases, order, model, budgets and scoring were unchanged. Course acceptance of the explicitly user-selected benchmark remains unverified. Earlier development experiments and the 17-call TritonAI seen-case workflow remain separate feasibility evidence. No email or assignment submission was sent.

Formal quality results

All 6 units completed. Strict pass counts over all 6 assigned units: clean 4/6; restart 3/6; selective 3/6; prevention 3/6; expiry-restart 2/6; expiry-selective 2/6; no-recovery 0/6. For numeric-port recovery, only 4 units were eligible: restart and selective each passed 3/4, with different failed units. The other 2 units remain not-injected, not quietly excluded. Collection is not a diagnosis endpoint. No-recovery fails the current-publication criterion by construction; 0/6 does not imply six incorrect diagnoses. Prevention checks source integrity, not arbitrary semantic truth.

Strict success requires diagnosed status, native joint root/target accuracy and evidence closure, input-supported reasoning, current publication, valid output contract and budget. Input-only judgments on all 36 produced answers were hash-sealed before native labels were accessed. Codex performed the review; it is not independent human validation. All 54 assigned arm slots remain in scoring. Provisional publication does not mean operational approval.

Development, costs and limits

Formal result and measurement boundary

Among four eligible formal units, selective recovery used 1 new call versus 3 for restart: 33,260 versus 64,902 incremental tokens (48.8% fewer). Equal 3/4 quality counts had different failed cases, not quality equivalence. None of four corrupted ports propagated. Both methods needed 3 calls on all six all-expired units. Formal timing sums API latency; full workflow latency and tracking overhead were not recorded.

Separate repeated handoff development: one seen case

handoff-dev-v2 ran three fresh prefixes on seen boutique/runtime/1 with R/S, S/R, R/S order. Specialists hand off summaries and a source catalog; the coordinator can retrieve 1-3 unchanged raw observations. A fixed summary edit changes httpGet to grpc. The raw-visible baseline differs in evidence exposure and call budget. All 21 arm slots remain, including three invalid no-recovery arms.

Paired recovery work, not total service savings

Restart/selective used 6/3 calls each repeat; tokens were 24,553/12,163, 25,825/12,176 and 25,660/12,020 (50.5%-53.2% fewer). Recovery-arm wall seconds were 14.319/6.640, 15.665/6.813 and 14.459/6.169, excluding notification, invalidation, cloning and final evidence persistence. Selective retained eight byte-identical valid artifacts per repeat; this alone does not establish semantic correctness.

Semantic evidence

Independent AI review was sealed before method mapping: all 15 core diagnoses were source-supported; corrupted handoff and raw-visible arms each adopted 0/3 false grpc claims. Both methods corrected the intermediate summary in 3/3 and restored six valid publications, but prior diagnoses were already correct. No-recovery rejected stale publication. Timing errors occurred in clean repeat 1 and selective repeats 1 and 3, none in restart. The reviewer had prior case context and is not human. Wrong-final-diagnosis repair and whole-answer equivalence remain unproven; one seen case supports no generalization claim.

Offline timing guard and recovery controls

Offline post-hoc replay flags 3/15 saved answers for timing conflict, 6/15 for review, and 6/15 with no detected conflict. All three known errors are flagged; this is retrospective development, not live correction or independent precision/recall. Five focused tests pass. Four synthetic seeded-state fixtures validate closure, stale-publication rejection and source refresh: restart/selective work units 6/3, 6/4, 6/6, 6/6. Handwritten recomputation proves no LLM semantic recovery. Nine tests pass in total.

Separate costs and next decision

Frozen formal: 92 calls, 519,696 tokens, $0.40225095 reconciled cost and no API errors. Separate handoff: 54 calls, 241,475 tokens, $0.2136544; settled course balance $49.11201275. Shared-prefix costs are excluded from paired recovery comparisons. Earlier causal-summary development stays separate. Retain nonpropagation under fixed faults; never select a fault because it causes failure.

Reproduction and timeline appendix

Portable evidence and immutable formal archive

The ZIP preserves formal traces/labels/hash seal and includes handoff raw inputs, outputs, costs, dependency records and blinded review. It adds timing-gate replay, synthetic controls, tests and preregistered protocol v3. Clean extraction verifies saved results without credentials or inference; installing pinned dependencies can require internet.

Offline verification after extraction

Create a Python >=3.11 environment and install requirements.lock. Run scripts/verify_formal_release.py for hashes, original review seal, 36 formal answers and 54 slots. Run summarize_handoff_saved.py on development/handoff-dev-v2. Re-run the timing and seeded-control scripts and compare JSON byte-for-byte; run the nine new tests. Exact commands are in docs/phase1-timing-reproduction-20261024.txt. Saved-output checks do not regenerate model responses or establish independent semantic validation.

Completed October 24

Frozen formal evaluation, native scoring, sealed evidence judgments, billing reconciliation and portable reproduction. Separate handoff development: three repeats on one seen case, fixed fault and order, all outcomes retained. AI assistance includes implementation, analysis and drafting. Independent AI review is distinct from student/human validation. No submission or operational action was performed.

October 24-November 4

Complete student review and exact live deadline/upload-slot verification. Saved overview lists November 4 for Phase 1. Benchmark course acceptance remains unverified. Timing guard is development-only: it proposes source-derived wording for review, does not silently rewrite old answers, and requires live generation calibration before a correction claim.

Next experimental milestone

Preregister separate natural-propagation and seeded conditional-recovery tracks; retain ineligible and nonpropagating outcomes. Require material ancillary correctness, source freshness and equal validation/repair budgets across methods. v3 proposes 6 timing-calibration plus at most 36 handoff calls; none executed. Verify model/key/price/balance and freeze an enforcing runner before spending. Next seen candidate is service/2; it is not an unseen test.

Remaining term and limitations

Analyze paired quality before resource savings. Multi-group scheduling and gated procedural learning follow the core recovery evidence. Prepare final artifacts/report/presentation in late November-December, checking official dates. Static snapshots, trusted invalidation and mutable model aliases limit claims. Sources: Cloud-OpsBench repository; saved course Option 5 overview and Piazza 46; TritonAI documentation.